

1. Introduction and Motivation

Early crack detection is an important aspect of infrastructure monitoring and preventive maintenance. Concrete structures are exposed to mechanical loads, temperature variations, humidity, and material aging, which can lead to the appearance and propagation of cracks.

Although cracks may initially appear harmless, they can be early indicators of structural deterioration. If left undetected, they may expand over time, increasing maintenance costs and potentially affecting users safety.

Traditional inspections rely on manual visual assessment, which can be time-consuming, costly, and dependent on the experience of the inspector. Computer Vision and Artificial Intelligence allow to automate this process, enabling faster, more consistent, and more scalable inspections.

2. Project Description

The objective of this project is to automatically classify concrete surface images into two categories:

- Crack
- Non-crack

Two approaches are investigated:

1. A traditional Computer Vision pipeline using HOG feature extraction and an SVM classifier.
2. A Deep Learning approach based on Convolutional Neural Networks (CNNs).

3. Dataset Description

The dataset was obtained from Kaggle and contains 40,000 concrete surface images:

- 20,000 crack images
- 20,000 non-crack images

The dataset is perfectly balanced, which helps prevent class bias during training. All images have a resolution of 227×227 pixels and exhibit relatively low levels of visual noise.

A Gaussian Blur filter was evaluated during preprocessing but did not produce noticeable improvements. Consequently, the final pipeline only relied on grayscale conversion before feature extraction.

4. Data Exploration

Several exploratory analyses were performed before model training:

- Verification of class distribution
- Visualization of sample images
- Verification of image dimensions
- Duplicate image inspection

The dataset was confirmed to be balanced, consistently sized, and free of significant duplication issues. These observations indicate that the data is suitable for classification tasks without extensive preprocessing.

5. Methodology

The traditional Computer Vision pipeline consists of four main stages:

1. Data Loading
2. Image Preprocessing
3. HOG Feature Extraction
4. SVM Classification

5.1 Data Loading

Images were loaded from the Positive and Negative directories and assigned binary labels:

- 1 for crack images
- 0 for non-crack images

5.2 Image Preprocessing

Since HOG relies on intensity variations rather than color information, all images were converted to grayscale.

5.3 HOG Feature Extraction

Histogram of Oriented Gradients (HOG) is a feature extraction technique that describes image structure through local gradient orientations.

For each image:

- Grayscale conversion
- Gradient computation
- Orientation histogram generation
- Feature vector construction

The resulting feature vectors were used as input for classification.

5.4 SVM Classification

The extracted HOG features were used to train a Support Vector Machine classifier.

The dataset was split into training and testing subsets using an 80/20 split to preserve class balance. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

6. Experimental Results

Several experiments were conducted using different dataset sizes and SVM configurations.

6.1 Linear SVM Results

Dataset Size	False Negatives	False Positives	Accuracy	Recall
500 images	6	0	94%	88%
2000 images	13	7	95%	94%
4000 images	20	7	96%	95%

As the dataset size increased, the model encountered a larger variety of crack patterns and textures. Although performance remained strong, the number of false negatives gradually increased.

Since false negatives correspond to missed cracks linked to the recall metric, minimizing this latter is particularly important for real-world inspection.

6.2 Hyperparameter Tuning

Several values of the regularization parameter C were tested (1, 2, 5, and 10).

Only negligible improvements were observed, suggesting that the main limitation was not the regularization strength but the linear nature of the decision boundary.

6.3 RBF SVM

To improve performance, a non-linear SVM using a Radial Basis Function (RBF) kernel was evaluated on the 4,000-image dataset.

Model	False Negatives	False Positives
Linear SVM	20	7
RBF SVM (C = 1)	9	2

The RBF kernel reduced false negatives by more than 50% while also decreasing false positives.

Performance metrics for the best model were:

- Accuracy: 98.62%
- Precision: 99%
- Recall: 98%
- F1-Score: 99%

Increasing C beyond 1 did not improve the results. Therefore, the best classical Computer Vision model obtained in this project was the RBF SVM with $C = 1$.

These results indicate that the crack and non-crack classes are not perfectly linearly separable and benefit from a more flexible decision boundary.

6.4 Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) was implemented to evaluate the performance of a Deep Learning approach on the concrete crack classification task.

the CNN approach eliminates the need for manual feature extraction by allowing the network to learn increasingly complex representations of cracks through multiple convolutional layers.

CNN Architecture

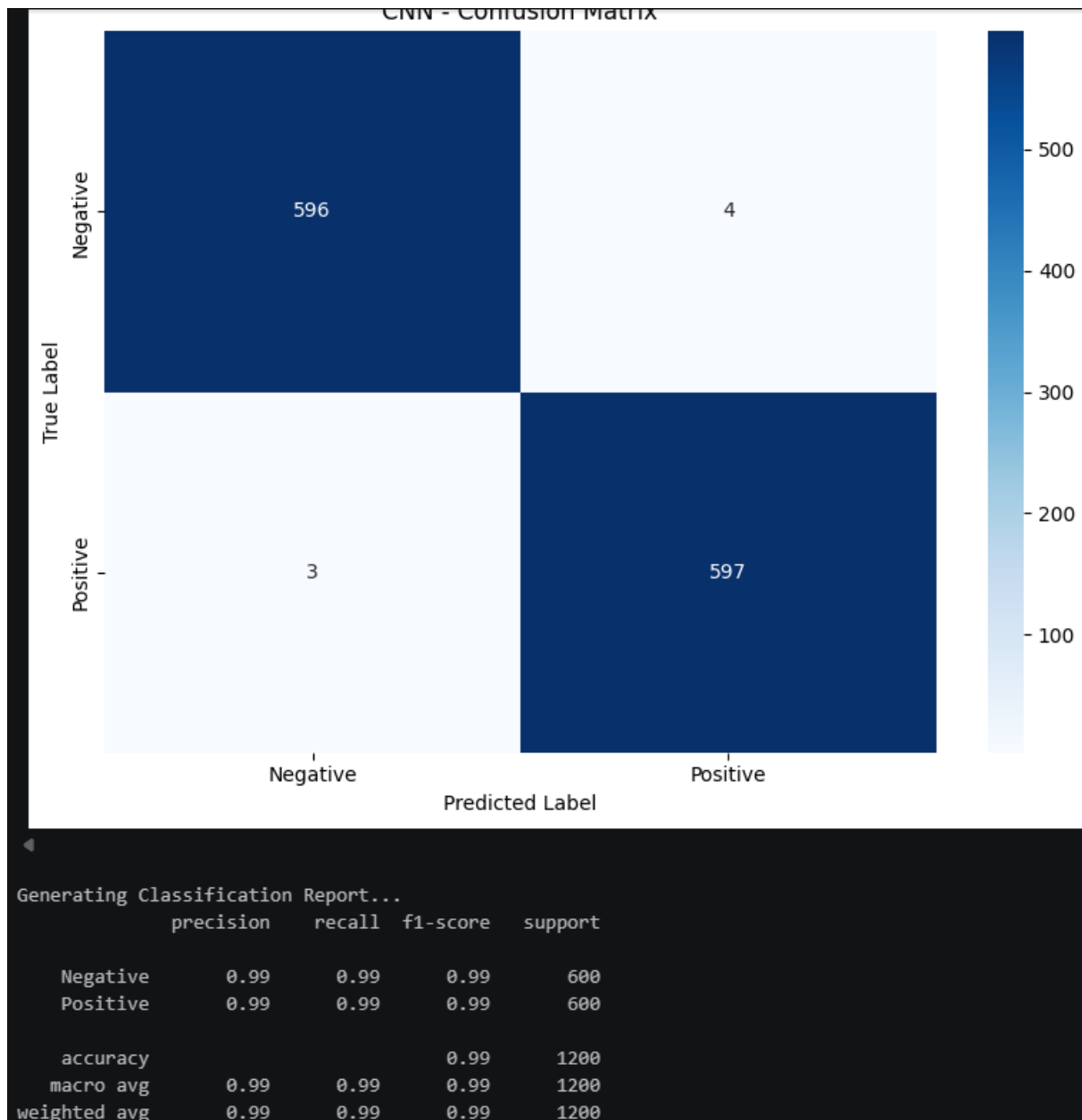
The implemented CNN consists of the following layers:

- Three convolutional blocks with 32, 64 and 128 filters respectively.
- A ReLU activation function after each convolution.
- A MaxPooling layer after each convolutional block to reduce spatial dimensions.
- A Flatten layer to convert the extracted feature maps into a one-dimensional feature vector.
- A fully connected Dense layer containing 128 neurons.
- A Dropout layer (0.5) to reduce overfitting.
- A final Dense layer with two neurons followed by a Softmax activation function for binary classification.

The model was trained using the Adam optimizer with the Binary Cross-Entropy loss function for 20 epochs.

CNN Experimental Results

The trained CNN achieved excellent classification performance on the test dataset.



The confusion matrix shows that only seven images were misclassified among the 1,200 testing samples.

- False Positives: 4
- False Negatives: 3

The final performance metrics were:

- Test Accuracy: 99.42%
- Precision: 99%
- Recall: 99%
- F1-Score: 99%

These results demonstrate that the CNN successfully learned discriminative crack features while maintaining a very low error rate.

Learning Curves Analysis

The learning curves show the training behavior of the CNN model. As shown in the accuracy curve, both the training and validation accuracy rapidly increase during the first epochs before stabilizing around 99%. The validation accuracy remains very close to the training accuracy during the entire training process, indicating that the model generalizes well to unseen data.

The loss curves show a similar behavior. Both the training and validation loss decrease significantly during the first epochs, showing that the optimization process successfully minimizes the prediction error. Although the validation loss fluctuates slightly after convergence, it remains low and stable, which is expected due to the variability of the validation batches.

Overall, the learning curves indicate that the CNN converges efficiently and does not exhibit significant overfitting.

6.5 Comparison Between Classical Computer Vision and CNN

Both approaches achieved excellent performance on the concrete crack dataset.

The HOG + SVM pipeline provided very competitive results for only few seconds of training. Due to HOG extracting features before classification, the SVM operates on compact feature vectors, resulting in a lightweight and computationally efficient solution.

In contrast, the CNN learns its own visual features directly from the images. Although the training process required approximately 20 epochs (around 16 minutes on my CPU), the CNN achieved the highest overall performance.

Model	Accuracy	False Negatives	False Positives	Training Time
Linear SVM	96.5%	20	7	~seconds
RBF SVM	98.62%	9	2	~9 seconds
CNN	99.42%	3	4	~16 minutes

The CNN reduced the total number of classification errors from 11 to 7 compared with the best classical approach.

However, this improvement came at the cost of a significantly longer training time on my basic hardware configuration (my CPU which is not the brighter light of the box) .

For applications where computational efficiency is important, the HOG + SVM pipeline remains an excellent solution.

For safety-critical inspection systems where maximizing crack detection accuracy is the primary objective, the CNN provides the best overall performance due to its lower number of missed cracks and its ability to learn image features automatically.

6.6 Failure Analysis

Despite both models achieved high performance, a small number of images were still misclassified.

SVM Failure Analysis

For the SVM models, the nature of the errors can be explained by the characteristics of the decision boundaries.

With the linear SVM, the model try separate the two classes using a single linear hyperplane. The linear SVM struggles to correctly classify samples that are near the decision boundary, leading to both false positives and false negatives. This might be due to the concrete crack pattern complexity.

The SVM with an RBF kernel improves performance by introducing non-linear decision boundaries. This allows the model to better capture complex patterns and variations in the data. The remaining errors are probably due to ambiguous images, such as very subtle cracks, noisy textures, or visual patterns that resemble cracks.

CNN Failure Analysis

The CNN achieved an accuracy of 99.42%, yet some errors remain.

Some crack images were predicted as non-crack (false negatives). These cases be due to very thin cracks or textures that resemble a lot to intact concrete surfaces.

Similarly, a few non-crack images were incorrectly classified as cracks (false positives). Such errors are probably caused by subtle cracks, noisy textures, or visual patterns that resemble cracks.

Overall, we can affirm that the low number of misclassified images shows that the CNN generalizes well and is robust to most variations present in the dataset.